THE SURROGATE GAP: WHEN DO LABEL-FREE BEHAVIORAL TESTS TRACK HUMAN AGREEMENT FOR LLM JUDGES?

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ABSTRACT

When a language-model judge scores AI-generated Lean 4 theorems and specifications against gold references, how much should we trust the judge itself? Revalidating every new judge, prompt, model, or threshold with fresh human ratings does not scale, so we study whether judge validation can be *amortized*: human labels are collected once, and new judge configurations are screened by automatic, falsifiable behavioral tests. We separate three objects prior drafts collapsed into one scalar: scoped judge *trustworthiness* TW_m (construct), its human-grounded *criterion* TI_{m_1} on a specified annotation panel, and a label-free behavioral *instrument* — a property vector $\mathbf{P}$ of identity, bug-monotonicity, specification-monotonicity, and stability probes with aggregate PTI_p. The instrument is a cheap *screen*: a probe violation is evidence against trustworthiness, while passing licenses acceptance only through a demonstrated *bridge* correlation with the criterion, within the validated judge population. On a fixed calibration panel (75 rated candidate–gold pairs, 72 analyzed; one annotator under two rubric-strictness views), the variance-aware aggregate $\mathrm{PTI}_{\mathrm{var}}$ tracks judge-level human agreement across three model families — validation-set Spearman 0.833 (averaged labels), 0.905 (strict), 0.738 (lenient) over $n{=}8$ configurations, with wide bootstrap intervals; all such numbers are calibration evidence, not held-out claims. We disclose two audited negatives. First, an independent five-rater pool (343 labels, 78 pairs) collected as a held-out test was annotated under a revised rubric and correlates with the calibration labels at only $\rho \approx 0.13$: the direct frozen test cannot answer the transfer question — criterion drift precedes power. Second, a pre-registered re-anchored screen over 32 judge configurations found no label-free property signal meeting its registered success bar. These results bound but do not settle label-free transfer, and they yield the protocol requirements judge validation needs: one rubric and one rater pool per claim, label-lineage checks, and screen semantics stated explicitly.

1 INTRODUCTION

Agentic AI systems increasingly generate both code and the trust artifacts meant to evaluate that code: tests, specifications, and formal theorems. In Lean 4, these artifacts are code-adjacent formal objects: runnable programs paired with theorem or specification statements and, sometimes, proofs. In classical autoformalization the target theorem is supplied, so evaluation can reduce to proof success; in agentic verification the system must decide *what should be true* before proving anything. The Lean kernel can check a proof, but it cannot say whether a proposed theorem is complete, semantically aligned with the intended program, or merely a weak statement that typechecks. As agents produce more candidate artifacts than humans can inspect, theorem-quality evaluation becomes a code-evaluation and scalable-oversight problem (Bowman et al., 2022; Zhou et al., 2025; Khattab et al., 2023; Cheng et al., 2024).

The standard reliability check is per-judge human validation: collect labels, measure agreement between judge and human ratings, and keep the judge if agreement is high. That check is necessary but scales linearly with the number of judges, prompts, model updates, thresholds, and domains; every new model family or prompt revision asks for another round of annotation. Humans remain the

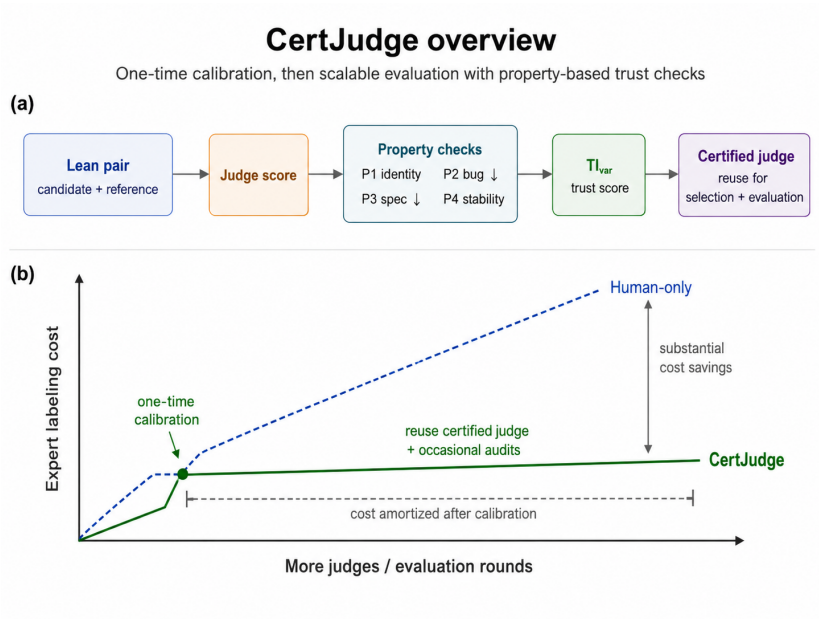


Figure 1: **Amortized validation of LLM judges.** A small human-labeled calibration panel is paired with falsifiable behavioral probes (P1–P4) to compute a label-free proxy trust index, enabling judge screening with reduced relabeling overhead. (The embedded figure text uses the legacy name $\mathrm{TI}_{\mathrm{var}}$ for the quantity we now write $\mathrm{PTI}_{\mathrm{var}}$; see Section 3.)

anchor — the waste is re-using them from scratch for every judge. Our thesis, in one sentence: *we test whether automatically measured behavioral properties can predict a scoped human-alignment criterion for unseen LLM judges, thereby amortizing — not eliminating — the cost of human validation.*

Following the validity-centered evaluation framework of Salaudeen et al. (2025), we separate the *construct* (scoped judge trustworthiness, TW_m), the *criterion* that measures it on a specified panel (a human-grounded index TI_m), and the *instrument* (a label-free property vector $\mathbf{P}$ with aggregate PTI_p). The properties are concrete: a judge should give maximal scores to identical artifacts, lower scores as semantic bugs accumulate, lower scores as specification elements are removed, and stable scores under repeats and semantics-preserving rewrites. The instrument has two jobs with different success metrics (Section 5): a *screen* that gates judges before humans are spent, and a *surrogate* that predicts the criterion numerically. Passing the screen never establishes trustworthiness by itself; acceptance is licensed only by a demonstrated bridge between PTI_p and TI_{m_1}, and only within the validated judge population.

Our **contributions** are:

1. We distinguish scoped judge trustworthiness TW_m (construct), its human-grounded criterion TI_m measured on a specified panel, and the automated behavioral instrument $\mathbf{P}(aj)$, making explicit which claims need human labels and which do not (Section 3).
2. We introduce a reusable suite of falsifiable behavioral probes for LLM judges of formal-code artifacts, with scripted perturbation ladders and pre-specified estimators (Section 4).
3. We evaluate whether the property aggregate is a valid surrogate for TI_{m_1}, separating property violations (screen) from prediction (surrogate). On the calibration panel the variance-aware aggregate tracks judge-level human agreement across three model families (validation-set Spearman $0.833/0.905/0.738$ on averaged/strict/lenient labels, $n{=}8$ configurations); we report uncertainty throughout and label every such number as validation-set evidence (Section 6).
4. We disclose the audited data facts and two negative results — criterion drift ($\rho \approx 0.13$ between labeling campaigns) that invalidates the direct frozen test, and a pre-registered 32-

configuration screen that found no qualifying label-free signal — and derive the protocol requirements they impose (Section 7).

2 Related Work

LLM-as-a-judge. LLMs are increasingly used as automated evaluators, with reliability commonly validated by agreement with people: G-Eval reports Spearman correlation with human scores (Liu et al., 2023), MT-Bench reports judge agreement with human preferences (Zheng et al., 2023), and Judge-Bench evaluates replication of human annotations across 20 NLP datasets (Bavaresco et al., 2024). Large-scale audits caution that reliability varies by task and setup and stress explicit validation before replacing human judgment; known biases include position, verbosity, and prompt sensitivity (Bavaresco et al., 2024; Li et al., 2024; Zheng et al., 2023). We keep human agreement as the criterion but amortize it: one calibration panel relates falsifiable behavioral properties to human agreement, after which new judge configurations are screened without new labels.

Improving and evaluating judge reliability. Debate-style panels (Chan et al., 2024), rubric-driven prompting (Liu et al., 2023), and internal-consistency measurement (Li et al., 2024) target reliability at inference time. For code, ICE-Score and CodeJudge highlight the difficulty of one-shot correctness assessment (Zhuo, 2024; Tong & Zhang, 2024). Consistency-focused analyses argue stability, not reasoning complexity, drives reliable grading (Chen et al., 2026). Our probes operationalize exactly this: identity, monotonicity under controlled edits, and stability under rewordings, in the spirit of metamorphic testing. Salaudeen et al. (2025) provide the construct/criterion/instrument vocabulary we adopt; Robertson & Koyejo (2025) give information-theoretic oversight queries for the verifier-absent regime, complementary to our verifier-present setting. Appendix K surveys scalable oversight, consensus aggregation (Denisov-Blanch et al., 2026), test-set-specialized autoraters (Finkelstein et al., 2025), autoformalization, and HCI perspectives.

3 Problem Setup and Definitions

Task. Each benchmark item contains a natural-language problem statement, a gold Lean 4 artifact g, and a candidate artifact x produced by an AI agent; artifacts may include code, tests or specifications, and a target theorem/specification statement. An LLM judge parameterized by $aj = (\text{model}, \text{prompt}, n_{\text{repeats}}, \text{aggregation})$ maps (x, g) to a score $\hat{s}_{aj}(x, g) \in [0, 1]$ reflecting how well the candidate captures the intended formal specification. Type-checking alone cannot decide this: a candidate may compile while being under-specified, over-constrained, vacuously true, or semantically shifted (Miranda et al., 2026), and no complete mechanical check of specification equivalence exists (Rice, 1953).

Three objects, three roles. Earlier versions of this work collapsed the following into one scalar; we keep them separate throughout (role labels follow Salaudeen et al., 2025):

Object	Role	Notation	Needs labels
Trustworthiness (predicate we care about)	construct	$\mathrm{TW}_m(aj)$	—
Trust index (criterion on a specified panel)	criterion	$\mathrm{TI}_m(aj)$	yes
Proxy trust index (behavioral aggregate)	instrument / surrogate	$\mathrm{PTI}_p(aj)$	no

Table 1: **Construct, criterion, instrument.** Subscripts follow the L_p convention: the capital letter names the family, the subscript selects the variant. m indexes the matching criterion on the target side; $p \subseteq \{1, \ldots, K\}$ indexes the property set on the proxy side and is locked before any bridge analysis (choosing p to maximize the bridge correlation is model selection over $2^K - 1$ subsets and reintroduces selection circularity).

The m-ladder. m_0: distributional match, $P_\theta(s \mid \cdot, aj) \approx P_H(s \mid \cdot, h_j)$. m_1: correlational — $\mathrm{Corr}(S_{aj}, S_{h_j})$ high over items. m_2: m_1 plus stability — $\mathrm{std}(\hat{s}_{aj})$ low. We write $\mathrm{TW}_2 \equiv \mathrm{TW}_{m_2}$; m_2 is the working target. The construct is

$$\begin{aligned}\mathrm{TW}_m(aj):\ &\forall a \in L_A,\ \text{if } aj \approx_\Pi h_j \text{ then} \\ &\left\| P_\theta(s \mid x, y^*, \hat{y}{=}a(x), aj) \approx_m P_H(s \mid x, y^*, \hat{y}{=}a(x), h_j) \right\|_m \text{ small},\end{aligned} \tag{1}$$

where $aj \approx_\Pi h_j$ is the *persona precondition*: the judge is conditioned toward the rater persona $h_j = (\text{person}, \pi)$, without which the definition is unsatisfiable over a heterogeneous rater pool. Items are what the m_1/m_2 criteria integrate over — they are never universally quantified in a correlational statement. The measurable objects are

$$\mathrm{TI}_{m_1}(aj) = \mathrm{Corr}_{i \in \text{panel}}\big(\hat{s}_{aj}(i),\, s^*_{h_j}(i)\big), \qquad \mathrm{PTI}_p(aj) = \mathrm{GM}\big(P_k(aj) : k \in p\big), \tag{2}$$

and at the m_2 target the criterion gains a stability conjunct,

$$\mathrm{TI}_{2'}(aj) = \Big(\mathrm{clip}_{[0,1]}\big(\mathrm{Corr}(D_{\hat{S}_{aj}}, D_{\hat{S}^*_H})\big) \cdot \big(1 - \mathrm{STD}[D_{\hat{S}_{aj}}]\big)\Big)^{1/2}, \tag{3}$$

with the stability term judge-side (it is P_4, the label-free conjunct), orientation-flipped, and the correlation clipped before the geometric mean. Correlations are computed only over *paired* observations — the same candidate scored by both sides — with repeats aggregated first: the K_a agent generations, K_{aj} judge repeats, and K_h human passes are three distinct indices that never align pairwise. Because one conjunct of TW_2 (stability) is directly estimable without labels via P_4 — an estimate with a confidence interval, not a verification — P_4 leaves the primary bridge aggregate: the headline bridge object going forward is $\mathrm{PTI}_{\{1,2,3\}}$ versus TI_{m_1}, the reported object is the profile $\big(\mathrm{PTI}_{\{1,2,3\}},\ P_4 \pm \mathrm{CI},\ \text{violation flags}\big)$, and the full variance-aware aggregate survives as a single-number deployment convenience (Section 4). Stability must never appear on both sides of a bridge.

Three claims. *Claim A* (definitional): aj is TW_2-trustworthy — needs humans. *Claim B* (proxy): $\mathrm{PTI}_p(aj) \geq \tau$ — label-free. *Claim C* (bridge): $\mathrm{PTI}_p \approx \mathrm{TI}_{m_1}$ over held-out judges — established once, population-scoped. Failing B is evidence against A with strength set by C; passing B never establishes A except through C. Refuting A from a single probe violation requires one of two repair routes. *Route A (definitional)*: define $\mathrm{TW}(aj) := [\mathrm{TI}_{m_1} \geq \tau_A] \wedge \bigwedge_k [P_k \geq \tau_k]$, so a probe violation rejects by definition and "properties predict alignment" is reported as a separate empirical result. *Route B (conditional falsification)*: under stated metamorphic-validity and margin assumptions, sufficiently large probe violations falsify pointwise ε-alignment (Lemma 1, Appendix B); the margins δ are currently assumed, not measured, and a small human collection on perturbed ladder items would ground the lemma while also testing whether ALPHA-renaming is human-invariant — the standing P_4 failure mode. We claim no "sound rejection by construction" at the ratified target: under TW_2, correlation is an aggregate, so a judge can carry high TI_{m_1} while exhibiting a local bug-ladder reversal, and P_k low does not logically imply TI_{m_1} low.

Amortization and its four generalization axes. The criterion is only measurable on the annotated subset $b' \subseteq b$; the proxy runs on the full benchmark: $\mathrm{PTI}(b, L_A, aj, P) \approx \mathrm{TI}_{2'}(b', L_A, aj, H)$. This exposes four axes that any amortization claim must name: held-out *judges* (the primary claim), new *agents*, new *benchmarks*, and within-benchmark *item-level extrapolation* $b' \to b$ — here the bridge is validated on 72 items while the proxy is used on 157×3 candidate artifacts. The honest cost claim is that labels are *de-indexed, not eliminated*:

$$C_{\text{amortized}} = \underbrace{|\Pi|\, n\, c_{\text{human}}}_{\text{persona conditioning}} + \underbrace{|AJ_{\text{val}}|\, |I|\, c_{\text{human}}}_{\text{bridge, once}} + \underbrace{|AJ|\, |A|\, |Q|\, c_{\text{API}}}_{\text{per screened judge}}, \tag{4}$$

which is $O(1)$ in agents and screened judges and $O(|\Pi|)$ in personas; approved phrasings are "no new human labels per screened judge" and "human-label amortized."

4 Behavioral Probes, Estimators, and Data Pipeline

The four properties. *P1 — identity & consistency*: the judge should assign the maximum score to an artifact compared with itself, $s(X, X) = 1$. *P2 — monotonicity to bugs*: scores should decrease as semantic bugs are cumulatively introduced. *P3 — monotonicity to missing specifications*: scores should decrease as theorems/tests are removed. *P4 — stability & semantic invariance*: scores should remain stable across repeated calls and under semantics-preserving rewrites; ALPHA-renaming changes only bound-variable names, so a robust judge should score original and renamed artifacts almost identically. We evaluate these with rank-based monotonicity statistics (Kendall's τ, Spearman's ρ) and pairwise ordering accuracy, following prior judge-evaluation work (Tong & Zhang, 2024; Zhuo, 2024).

Three datasets, one type rule. Items pair *inside* each side; judges pair *across* sides. (1) *Probe-response tensor* (the instrument's data; label-free): $D_Q(aj)[f, T, r, k] = \hat{s}_{aj}(T_r(x_f), g_f)$ over base artifacts $f \in F$ (the published run: $|F| = 184$), transform families T (identity-self; RENAME/REORDER/STYLE/ALPHA; bug ladder L0–L4; cumulative spec-removal ladder), rung r, and repeats $k \leq K_{aj} = 3$. This tensor has *no human axis*: the supervision labels provide on the gold side is provided here by construction — the ladder order is known because we built it. The suite is scripted once from existing candidates and golds, cached, and reused across judges; per new judge the cost is inference only, zero annotation. (2) *Panel scores* (the criterion's judge side): the judge's repeat-aggregated scores on the 72 aligned pairs — required only for judges in the bridge set, never for newly screened judges. (3) *Human tensor*: 72 items $\times$ {strict, lenient, avg}; fixed, collected once. Estimators collapse (1) to the property vector — e.g. a ladder property is $\frac{1}{|F|}\sum_f \mathrm{clip}\big(\frac{1-\rho_f}{2}, 0, 1\big)$ with ρ_f the Spearman correlation of (rung, median-over-repeats score); P_4 alone consumes the un-aggregated repeat axis — and (2)+(3) collapse to $\mathrm{TI}_{m_1}(aj) = \mathrm{Spearman}_{i\in 72}(\mathrm{median}_k\, \hat{s}_{aj}(i), s^*_H(i))$ per label view. The *bridge dataset is judge-indexed*: $\{(\mathrm{PTI}_p(aj_j), \mathrm{TI}_{m_1}(aj_j))\}_{j=1}^n$, one point per judge configuration ($n = 8$–13 today, target ≈ 40), with $\hat{\rho}$ bootstrapped over judges. D_Q and the human tensor share no index and are never correlated row-wise — humans never rated perturbed variants, by design. This is why the outer power constraint counts *judges*, not items, and why the fix is more judge configurations.

Aggregates. The measured aggregate in this study's runs is the variance-aware

$$\mathrm{PTI}_{\mathrm{var}} = \exp\Big(\tfrac{1}{3}\log C_f + \tfrac{1}{3}\log(1 - \widetilde{\mathrm{VarCorr}}) + \tfrac{1}{3}\log(1 - \widetilde{\mathrm{VarScore}})\Big), \tag{5}$$

where C_f is the geometric aggregate of the four property scores and $\widetilde{\mathrm{VarCorr}}, \widetilde{\mathrm{VarScore}}$ are normalized variance penalties (Appendix C); simple arithmetic variants $v1$/$v2$ serve as comparators. Because the criterion measured in this paper is TI_{m_1} (correlation only, no stability on the target side), reporting $\mathrm{PTI}_{\mathrm{var}}$ against TI_{m_1} does not place stability on both sides; at the m_2 target, the primary bridge must be $\mathrm{PTI}_{\{1,2,3\}}$ with P_4 reported standalone. *The geometric mean is a soft AND, not a gate*: $\mathrm{GM}(1, 1, 1, 0.5) \approx 0.84$, so the aggregate collapses only at exactly zero. It remains the right ranking summary (zero-annihilating, imbalance-penalizing), but the hard screening decision uses per-property confidence bounds — FAIL-SCREEN when $\mathrm{UCB}(P_k) < \tau_k$ for any required k — and the auditable object is always the property vector $\mathbf{P}(aj)$, with $\min_k P_k$ reported when worst-case conformance matters.

The SCREEN workflow. (1) Label a fixed panel once; (2) evaluate P1–P4 for bridge-set judges on scripted perturbations; (3) establish the bridge with uncertainty; (4) screen new configurations by property checks alone — FAIL-SCREEN on a per-property violation, PASS-SCREEN licenses acceptance only through the bridge within the validated population, DEFER otherwise.

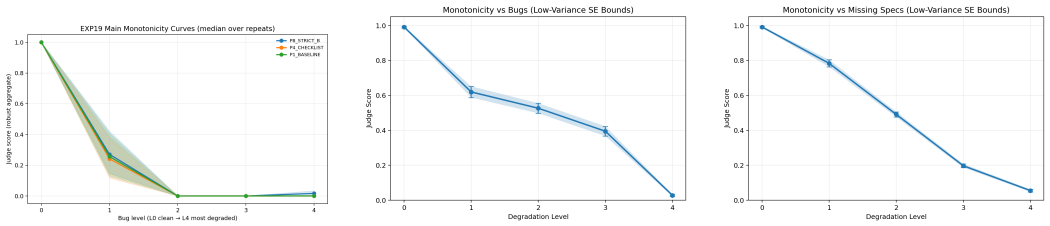
5 TWO JOBS, TWO VALIDATIONS

The instrument does two jobs that require different evidence, and the paper never conflates their metrics.

	Screen (gate before humans)	**Surrogate (predict the criterion)**
Claim	catches unacceptable judges cheaply	PTI_p tracks TI_{m_1} numerically
Decision	FAIL-SCREEN / PASS-SCREEN / DEFER	predicted score with interval
Metrics	false-pass rate, false-reject rate, defer rate, coverage, human ratings saved at a fixed false-pass tolerance	held-out rank correlation, MAE/calibration, prediction intervals, transfer across families/prompts/personas/agents

Table 2: **Screen vs. surrogate.** Decision utility — *at false-pass tolerance α, the screen avoids X% of human validations* — is the headline scalable-oversight quantity; measuring it on an expanded judge panel is registered as the next analysis and no value is claimed here.

For power-limited analyses we pre-register a smallest effect size of interest (SESOI; candidate $\rho_0 = 0.5$, the level below which screening saves less than it costs) and run equivalence tests, so an



(a) Average score under increasing degradation

(b) P2: Monotonicity vs. bugs (mean ± SE)

(c) P3: Monotonicity vs. missing specs (mean ± SE)

Figure 2: **Judge behavioral sanity checks.** VERIBENCH-wide GPT-5.3-Codex API run ($n_{\text{files}} = 184$, $n_{\text{repeats}} = 3$). **(a)** Smooth non-increasing score behavior under controlled degradations. **(b)** The P2 bug-ladder trend is directionally correct with local reversals (Pearson $r = -0.687$, Spearman $\rho = -0.808$, weak monotonicity 52.2%). **(c)** The P3 missing-spec trend is stronger and more stable (Pearson $r = -0.892$, Spearman $\rho = -0.899$, weak monotonicity 89.1%). Strict P1 identity and P4 rewrite invariance are reported in Table 3.

inconclusive interval is reported as inconclusive and a bounded null becomes a finding. A learned calibration map $f_{\text{cal}} : \mathbf{P} \mapsto \text{TI}_{m_1}$ is deferred until roughly 40 judge configurations exist; at $n \approx 8$–13 the honest object is the pre-specified aggregate with its raw correlation.

6 EXPERIMENTS: VALIDATION-SET BRIDGE EVIDENCE

6.1 DATA AND HUMAN-ANNOTATION PROTOCOL (AUDITED)

All experiments use VERIBENCH (Miranda et al., 2026), which comprises 857 theorems over five subsets: **HumanEval** (387 theorems from 56 programming puzzles Chen et al., 2021), **Easy** (278 from 41 introductory tasks), **CS** (68 from 10 classical algorithm problems), **Security** (112 from 28 vulnerability labs), and **RealCode** (12 from 5 Python standard-library programs).

Calibration panel. Judge validation uses an aligned panel of **75 rated candidate–gold pairs** = 25 problems × 3 agent generations (generations by Codex-, Claude-, and Llama-family agents); **72 pairs (24 problems) enter all analyses**, because one problem's gold artifact is indexed under a pre-renumbering filename and is skipped by the gold lookup. A repository audit reproduced every published statistic from the raw label files; Appendix A gives the reconciliation, and the supplementary archive contains the tidy panel. **One annotator, two rubric-strictness rounds**: a single expert annotator scored every pair twice on a 0–5 rubric under a *strict* and a *lenient* reading (150 rated labels; 144 analyzed). We refer to these throughout as the strict and lenient views (they were called `pass1`/`pass2` in earlier versions of this work; those names are inverted relative to chronology, so we avoid temporal language entirely). Inter-round agreement on the 72 analyzed pairs is moderate: Spearman 0.249 (95% confidence interval $[-0.02, 0.49]$), Pearson 0.375, quadratic-weighted $\kappa = 0.372$, MAE 0.771. With one rater, this is *rubric stability*, not inter-rater agreement: the criterion itself moves this much under a strictness change alone, which is why we report strict, lenient, and averaged label views side by side and treat single-view correlations as bounded by this ceiling. **Item selection.** The panel is stratified 5-per-subset across the five VERIBENCH subsets but *hand-picked* within subsets, frozen before any judge experiments, with no selection script or seed and no compile filtering; the estimand is therefore alignment on a fixed curated panel, not on a probability sample of the benchmark (a seeded stratified-selection template exists for a successor study and is the upgrade path if a probability sample is demanded). The same panel is reused across judges, so differences in trend reflect judge behavior rather than row selection.

6.2 PROPERTY SANITY CHECKS

Figure 2 shows the Codex judge satisfies the intended qualitative behavior: scores decrease as bugs accumulate and as specifications are removed. The strongest signal is specification-ablation monotonicity ($\rho = -0.899$); bug monotonicity is directionally correct but noisier ($\rho = -0.808$), with occasional local reversals. The same monotone pattern holds for Claude Opus 4.7 under matched plotting protocols (Appendix H).

Identity variant	Mean score	Mean within-item std
RENAME	0.936	0.005
REORDER	0.849	0.029
STYLE	0.982	0.001
ALPHA	0.399	0.096

Table 3: **P4 semantic-invariance breakdown (Codex API run).** The weakest mode is aggressive ALPHA-renaming — the standing P_4 failure mode (Section 8).

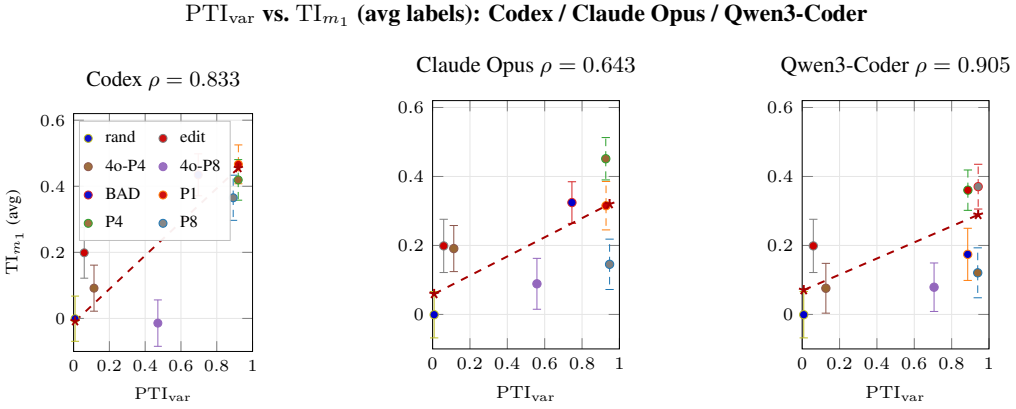


Figure 3: **Judge-level bridge:** $\mathrm{PTI}_{\mathrm{var}}$ **vs.** TI_{m_1}**.** Each point is one judge configuration; the y-axis is Spearman agreement with averaged human labels on the analyzed panel (n=72 rows). Vertical bars show approximate bootstrap standard errors; dashed lines are OLS guides. Across families, higher aggregate property satisfaction tracks higher human agreement. All values are validation-set calibration evidence (Section 7).

6.3 THE BRIDGE: PROPERTY AGGREGATE VS. JUDGE-LEVEL HUMAN AGREEMENT

The bridge question is whether the label-free aggregate tracks TI_{m_1} over judge configurations. Across Codex, Claude Opus, and Qwen3-Coder, judges with higher $\mathrm{PTI}_{\mathrm{var}}$ also agree more with human ratings. Figure 3 plots judge-level $\mathrm{PTI}_{\mathrm{var}}$ against TI_{m_1} (averaged labels) for the three families, holding fixed the 72 analyzed panel rows within each regression; Table 4 reports all three label views. $\mathrm{PTI}_{\mathrm{var}}$ does not only reward average property satisfaction — it also penalizes instability across repeats, prompts, and perturbations.

Table 4 reports the same relationship for strict, lenient, and averaged labels; six of nine family/view combinations exceed 0.7. Uncertainty at n=8 configurations is substantial and we report it wherever a headline number appears: for the Codex panel, 10k-resample bootstrap 95% confidence intervals are $[0.24, 1.00]$ (avg), $[0.41, 1.00]$ (strict), and $[0.11, 1.00]$ (lenient), with two-sided permutation $p = 0.017/0.005/0.045$ respectively; Pearson shows the same pattern (avg $r = 0.774$, strict 0.661, lenient 0.837), and Kendall τ is noisier (0.571/0.571/0.643). The intervals are wide, as expected at this panel size, but all exclude zero. The outer n here is the number of judge configurations — the item count cannot fix it (Section 4) — so expanding toward $\approx$40 configurations is the registered path to a powered bridge estimate.

Which aggregate captures the trend? Relative to arithmetic baselines, $\mathrm{PTI}_{\mathrm{var}}$ uniformly improves over $v2$ for every family: Codex $0.476 \rightarrow 0.833$, Claude Opus $0.143 \rightarrow 0.643$, Qwen3-Coder $0.500 \rightarrow 0.905$ (avg labels). The plain geometric aggregate is competitive for Codex and Opus (0.857 and 0.714), while $\mathrm{PTI}_{\mathrm{var}}$ is most consistently strong across families (Table 5); an expanded 13-judge panel stays above 0.75 across label views (Appendix E). We optimize for cross-judge consistency rather than single-family peak correlation.

Per-property signal. For averaged labels, P3 specification-ablation monotonicity is the strongest single predictor (Spearman 0.747, $p = 0.043$; Pearson 0.744, $p = 0.048$), with P2 bug-rank monotonicity next (Spearman 0.714, $p = 0.056$; Pearson 0.636, $p = 0.037$); P1 and P4 are weaker standalone predictors at $n_{\text{judges}} = 8$. This ordering is consistent across all three label views, indi-

Model family	strict	lenient	avg
GPT-5.3-Codex	0.905	0.738	0.833
Claude Opus 4.7	0.810	0.595	0.643
Qwen3-Coder	0.476	0.905	0.905

Table 4: **Judge-level bridge by label view.** Spearman correlation between judge-level $\mathrm{PTI}_{\mathrm{var}}$ and judge-level TI_{m_1} on the analyzed panel (n=8 configurations per family). View-specific values differ with rubric severity; the qualitative pattern is stable across families.

Model	v1	v2	GM	$\mathrm{PTI}_{\mathrm{var}}$
Codex	0.762	0.476	0.857	0.833
Claude Opus	0.429	0.143	0.714	0.643
Qwen3-Coder	0.524	0.500	0.786	0.905

Table 5: **Avg-label judge-level trends by aggregate formulation.** Spearman correlation between aggregate score and per-judge TI_{m_1} on averaged labels. $v1$/$v2$: arithmetic variants; GM: geometric monotonicity/stability; $\mathrm{PTI}_{\mathrm{var}}$: variance-aware (Eq. 5).

cating that semantic monotone behavior under controlled degradations is the most stable predictor of validation-set human agreement in this study. Notably, the intentionally bad lexical baseline BAD_STRINGY achieves moderate validation-set human Spearman (≈ 0.45) while failing weak monotonicity (12.5%): aggregate human agreement alone does not detect this judge's failure, but the property vector does — the screen catches what the criterion's aggregate misses.

Cross-view robustness. When the aggregate's weighting is tuned on one label view and evaluated on the other, the judge-level Spearman stays above 0.7 in both directions (strict→lenient: fit 0.881, test 0.810; lenient→strict: fit 0.857, test 0.786).

Decision-threshold calibration and downstream use. Converting scores to yes/no decisions at threshold $t \in \{0.60, \ldots, 0.85\}$ leaves ranking behavior unchanged ($\rho_{\text{bug}} = -0.790$ throughout) while stricter thresholds modestly reduce decision flips at some coverage cost; we select $t = 0.85$ (flip rate 0.046, pass rate 0.536; full sweep in Appendix F). As a downstream portability check — outside the eight-judge bridge panel — a property-checked judge ranks VERIBENCH theorem-generation agents in the expected order (baseline prompting 0.470; DSPy ReAct 0.615; Trace++ 0.588; Appendix G); this is a sanity check, not independent proof of judge correctness.

7 CRITERION DRIFT AND THE PRE-REGISTERED HELD-OUT OUTCOME

Everything above is validation-set evidence. The same human labels were used to choose the aggregate form, repeat aggregation, threshold, and prompt panel, so Section 6's trends are calibration findings, not out-of-sample claims. This section reports what happened when we tested beyond them.

The held-out pool measures a different criterion. Beyond the calibration panel, an independent pool was collected as the intended held-out test: five raters, 343 scored labels over 78 distinct candidate–gold pairs, many pairs with 4–5 independent ratings. The pool was annotated under a *revised* rubric protocol, and its labels are nearly uncorrelated with the calibration labels on shared items (Spearman ≈ 0.13 on 63 shared pairs). A bridge calibrated on criterion A and tested on criterion B with $\mathrm{Corr}(A, B) \approx 0.13$ fails at any judge count, so the direct frozen transfer test cannot answer the transfer question, and we withhold its correlations as uninformative rather than negative. *Criterion drift precedes power.* The pool itself is reliable within its own protocol: its estimated pool-mean reliability ceiling is ≈ 0.87, and an annotator re-labeling under the revised protocol matches the pool mean at Spearman 0.90 — label noise cannot be blamed for a null in a protocol-consistent rerun.

Pre-registered screens. Two pre-registered analyses asked whether any label-free property signal predicts judge-level alignment with the revised-protocol criterion. The first (14 configurations) returned its registered "underpowered" outcome. The second — expanded to 32 judge configurations,

with candidates, decision thresholds, and analysis seed committed before the new judge scores existed — found no candidate meeting the registered success bar (strongest candidate $|\rho| = 0.39$, Holm-corrected $p = 0.25$; leave-one-out selection transfer $\rho = 0.24$, permutation $p = 0.49$): a registered negative for label-free transfer at this power.

Both sample sizes, both intervals. Power remains binding on top of drift: the outer (bridge) correlation's n is the number of judge configurations (8–13 today; confidence intervals roughly ± 0.5), while the inner correlation runs over 72 items (intervals roughly ± 0.2). We report both, and the fix for the outer constraint is more judge configurations (≈ 40, API-priced), not more items.

What this establishes. A protocol requirement rather than a verdict on the method: one labeling protocol end-to-end — re-anchoring calibration to the revised-protocol criterion before any frozen test — with one rubric and one rater pool per claim, and label-lineage verification alongside the behavioral properties. The rubric is part of the criterion's identity; with $|H| = 1$ the panel criterion is one person's judgment under a stated rubric — legitimate as a *scoped* criterion via the $h_j = (\text{person}, \pi)$ decomposition, but it must be said plainly, and we say it here.

8 Failure Modes and Limitations

ALPHA-renaming. Among semantics-preserving rewrites, aggressive ALPHA-renaming is systematically the hardest for our judges (mean score 0.399 vs. ≥ 0.849 for milder edits; Table 3). We interpret this as a mismatch between what the kernel guarantees and what LLM judges implicitly condition on: tokenized programs under ALPHA sit far from both the gold reference and typical training text, so lexical cues shift scores even when equivalence is obvious to an expert. Excluding ALPHA variants materially changes measured robustness (trend-$v1$ Spearman: strict $0.810 \to 0.690$, lenient $0.810 \to 0.714$, avg $0.881 \to 0.786$), and local bug-ladder reversals occur despite strong global trends (per-judge reversal rates in Appendix D). A post-hoc canonicalization proxy reduces mean within-item dispersion under ALPHA from 0.250 to 0.070 (median per-item reduction 64%); pre-scoring canonicalization requires rescoring and is future work. Whether ALPHA-renaming is human-invariant on these artifacts is exactly the margin measurement Route B needs (Appendix B).

Limitations. The judge is a proxy: scores are prompt- and rubric-sensitive and do not constitute kernel-level checking; evaluations are specification-relative (Appendix J). No finite probe suite certifies all failure modes; untested modes may persist. The calibration panel is small, single-rater, and hand-curated, so the criterion is scoped to that panel and rubric; the bridge sample is n=8–13 judge configurations with wide intervals; all reported correlations are validation-set numbers; and the pre-registered screens returned underpowered and negative outcomes on a drifted criterion — cross-protocol transfer is bounded, not settled. Item-level extrapolation ($b' \to b$: 72 validated items vs. 157×3 scored candidates) is a stated assumption, not a measured one.

9 Conclusion

We reframed judge validation as measurement: a construct (TW_m), a human-grounded criterion (TI_m) on a specified panel, and a label-free instrument ($\mathbf{P}$, PTI_p) whose acceptance semantics are those of a screen backed by a bridge. On the calibration panel the bridge is strong and consistent across three model families; on a drifted criterion, a pre-registered screen found no qualifying label-free signal, and the audit of our own data shows how much of "ground truth" is protocol. The decisive next steps are mechanical: one rubric end-to-end, ≈ 40 judge configurations for a powered bridge, per-property screening bounds, margin measurement for Route B, and decision-utility measurement — human validations avoided at a fixed false-pass tolerance — as the headline oversight quantity.

Reproducibility Statement

The supplementary material contains an anonymized archive with: the tidy calibration panel (75 rated / 72 analyzed pairs with strict, lenient, and averaged views), the audit scripts that recompute every data fact in Section 6.1 from the raw label files (pure standard-library Python), the probe-suite definitions and judge prompt templates (Appendix I), cached per-judge property and bridge statistics

behind every table, and the pre-registration documents for the two held-out screens. Appendix A reconciles every published count and statistic against the raw files, including the two corrections this version makes (the 75-vs-72 panel description and the duplicate-handling convention in MAE/MSE).

THE USE OF LARGE LANGUAGE MODELS

Large language models were used in two capacities. (i) As research subjects: all judges under study are LLMs, as described throughout. (ii) As assistants: LLM-based coding tools were used for editing, LaTeX porting, repository auditing, and cross-checking numbers against cached experiment artifacts, under author direction and review. All claims, analyses, and final text were verified by the authors; the authors take full responsibility for the content.

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A AUDITED DATA FACTS AND COUNTS RECONCILIATION

This appendix states the audited facts about the human-annotation data that the main text relies on; the audit scripts and tidy panel ship in the supplementary archive.

Panel composition. The label files hold **75 rated candidate–gold pairs over 25 problems** (25 problems $\times$ 3 agent generations; 5 problems per VERIBENCH subset $\times$ 5 subsets $\times$ 3 generations $= 75$). Every published judge-alignment and inter-round number is computed on **72 pairs over 24 problems**: one problem's rows carry a pre-renumbering filename, the gold lookup misses, and its 3 pairs are skipped as `missing_gold`. The dropped rows are a stale *key*, not missing data — the candidate artifacts exist on disk under the renumbered name — so the two repair options are (a) this paper's wording (75 rated / 72 analyzed) or (b) renaming the 3 label rows and regenerating all dependent statistics; we adopt (a) and register (b) as a data-repair task.

The two unrelated "3"s. The 75-vs-72 gap (3 dropped pairs) is distinct from a second legacy artifact: in the *lenient* label file one problem appears twice (all 3 of its candidates, scored 1 and 2). Those duplicate rows collapse to the same 75 pairs and never inflated the panel; averaging them produces the three 1.5 scores visible in Figure 12.

Label totals and views. One expert annotator, two rubric-strictness rounds: 150 rated labels (75×2), 144 analyzed (72×2), on a 0–5 rubric, plus a derived averaged view. The strict and lenient names replace the earlier `pass1`/`pass2`, whose numbering is inverted relative to chronology (verified by content hash against the raw files); no per-row rater field exists anywhere in the label data, so rater identity is carried by file provenance only.

Inter-round agreement, one convention. On the 72 analyzed pairs, with duplicate rows averaged before comparison (mean-of-duplicates, the same convention as the averaged view): Spearman 0.249 (95% CI $[-0.015, 0.489]$), Pearson 0.375 ($[0.076, 0.604]$), Kendall τ 0.221, quadratic-weighted κ 0.372 ($[0.073, 0.581]$), MAE 0.771, MSE 1.122. An earlier version reported MAE 0.778 and MSE 1.139, which mix a first-of-duplicates convention into the same table; the values above are self-consistent under one convention. With one rater these statistics are *rubric stability*, not inter-rater agreement.

Item selection provenance. Stratified 5-per-subset, hand-picked within subset, frozen before any judge experiments; no selection script, no seed, no compile filtering. The estimand is alignment on a fixed curated panel. A seeded, audited stratified-selection template (30 problems, fixed seed, per-rater sheets) exists for a successor study and is the upgrade path to a probability sample.

The independent five-rater pool. 343 labels over 78 distinct pairs from 5 raters under a revised rubric; overlap with the calibration panel is 63 pairs; Spearman between pool mean and calibration labels on shared items is ≈ 0.13. The two label populations are not interchangeable, and any experiment that calibrates on one and tests on the other is confounded by construct drift (Section 7).

Agents on the panel. The three generations per problem come from Codex-, Claude-, and Llama-family agents; the full candidate pool spans 471 files $= 157$ problems $\times 3$ generations (problem granularity; the benchmark's 857 counts theorems).

B ROUTE B: CONDITIONAL FALSIFICATION LEMMA

Lemma 1 (Conditional falsification)**.** *Suppose the judge is pointwise ε-aligned on a class Z: $|J(z) - H(z)| \le \varepsilon$ for all $z \in Z$. Let T be a transformation with human margin $H(x) - H(Tx) \ge \delta$ for $x, Tx \in Z$. Then $J(x) - J(Tx) \ge \delta - 2\varepsilon$. Consequently, an observed reversal $J(Tx) - J(x) > 2\varepsilon - \delta$ refutes pointwise ε-alignment on Z; and if T is human-invariant ($\delta = 0$ with equality), then $|J(x) - J(Tx)| \le 2\varepsilon$.*

Proof. $J(x) \ge H(x) - \varepsilon \ge H(Tx) + \delta - \varepsilon \ge J(Tx) + \delta - 2\varepsilon$. □

The paper-level sentence licensed by this lemma: *under stated metamorphic-validity and margin assumptions, sufficiently large probe violations falsify pointwise ε-alignment.* The margins δ are currently assumed, not measured. A small human collection on perturbed ladder items (a few dozen ratings on bug-ladder rungs and ALPHA pairs) would (i) ground the lemma's premises empirically

and (ii) adjudicate whether ALPHA-renaming is human-invariant on these artifacts, which is the standing P_4 failure mode. This is registered as the δ-measurement collection.

C AGGREGATE (INDEX) DEFINITIONS AND COMPARATORS

The arithmetic baselines are

$$v1 = 0.30\,\text{ident} + 0.30\,\text{mono_rank} + 0.25\,\text{weak} + 0.15\,\text{stab},$$

$$v2 = 0.20\,\text{ident} + 0.20\,\text{mono_rank} + 0.15\,\text{weak} + 0.10\,\text{stab} + 0.20\,\text{anti} + 0.15\,\text{subtle},$$

where $\text{mono_rank} = (1 - \rho_{\text{bug}})/2$ rescales bug-ladder Spearman so larger is better, and $\text{stab} = 1 - \text{flip_rate}$. For the variance-aware aggregate (Eq. 5), the quality term is

$$C_f = (\text{ident} \cdot \text{mono_rank} \cdot \text{weak} \cdot \text{stab})^{1/4},$$

and $\widetilde{\text{VarCorr}}, \widetilde{\text{VarScore}}$ are normalized variance penalties (lower is better before $1 - \cdot$). The plain geometric comparator used in Table 5 is $\text{GM} = \text{mono}^{0.4} \cdot \text{stab}^{0.6}$.

Leanstral negative control under re-aggregation. To stress-test whether aggregate choice alone can rescue a weak judge family, we re-aggregate the same Leanstral judge outputs under AM ($v2$), GM, and PTI_{var}:

Label view	AM (v2)	GM	PTI_{var}
strict	-0.5952	-0.1429	-0.1190
lenient	-0.1905	-0.2619	-0.0476
avg	-0.4286	-0.2143	-0.0952

Table 6: **Leanstral remains low-alignment under multiple aggregate constructions.** Re-aggregation improves over the arithmetic baseline in some views, but all settings remain near-zero or negative, suggesting the bottleneck is the underlying judge behavior for this pairwise-rubric task rather than aggregate form. One testable conjecture: Leanstral is stronger at Lean proof editing/generation than at calibrated pairwise judging under this rubric.

D PER-PROPERTY DIAGNOSTICS AND FAILURE MODES

Reversal rates. Local bug-ladder reversal rates by judge, from the low-variance replot package: GPT-5.3 P4/P1 $= 0.00$, GPT-5.3 P8 $= 0.167$, BAD_STRINGY $= 0.833$, GPT-4o variants $= 0.667$–0.750, edit-distance/random $= 1.00$ (Figure 5).

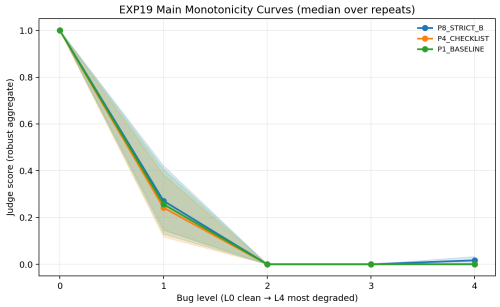
ALPHA isolation. Figure 4 separates the main monotonicity trend from the ALPHA-renaming failure mode rather than mixing them in one curve. Excluding ALPHA variants changes trend-$v1$ Spearman: strict $0.810 \to 0.690$, lenient $0.810 \to 0.714$, avg $0.881 \to 0.786$.

E VALIDATION-SET BRIDGE: FULL DIAGNOSTICS

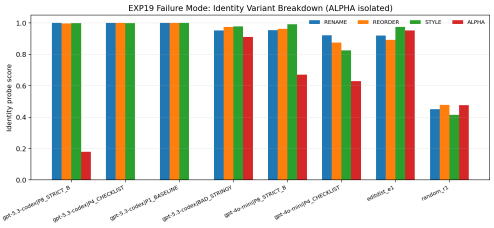
This appendix reports full visual diagnostics for the validation-set bridge analyses under the three label views (strict, lenient, avg); several embedded figure legends predate the strict/lenient rename and show `pass1`/`pass2` (strict = `pass1`, lenient = `pass2`). We evaluate eight judge configurations spanning frontier-model prompts, weaker LLM baselines, edit-distance heuristics, and random controls. 10,000-resample bootstrap 95% confidence intervals for the headline bridge coefficients are $[0.24, 1.00]$ (avg), $[0.41, 1.00]$ (strict), and $[0.11, 1.00]$ (lenient), with two-sided permutation p-values 0.017, 0.005, and 0.045 (fixed seed 42); the intervals are wide, as expected at this panel size, but all exclude zero.

E.1 LATEST RERUN (HIGHER REPEATS, ROBUST AGGREGATION)

To reduce judge-call noise, the multi-judge validation trend was rerun with `JUDGE_REPEATS=9`, median repeat aggregation, 10k bootstrap and 20k permutation resamples:



(a) Main low-variance monotonicity curves (core GPT-5.3 judges, robust repeat aggregation).



(b) Failure-mode isolation: identity-variant breakdown with ALPHA shown separately.

Figure 4: **Low-variance replot package: cleaner main trend + explicit failure separation.**

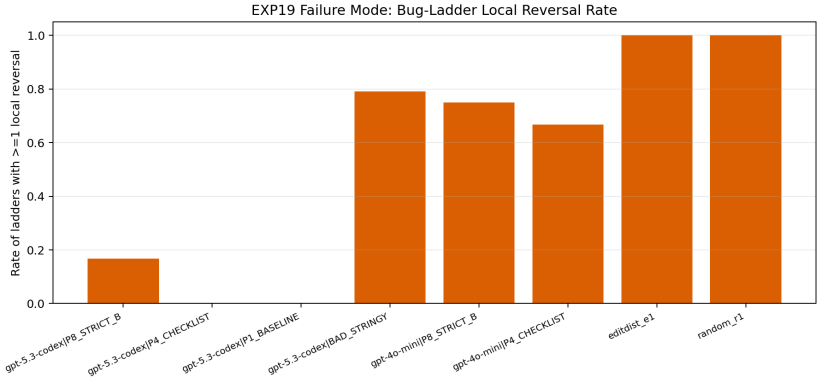


Figure 5: **Local bug-ladder reversal rates by judge.** Surfaces where monotonicity fails locally, separating robustness diagnostics from the primary trend plot.

E.2 Naive blind baseline

A single-pass, surface-form-oriented blind rubric without property calibration tracks lenient labels somewhat but remains substantially weaker and less consistent than the top-ranked judges:

E.3 Human-agreement ceiling context

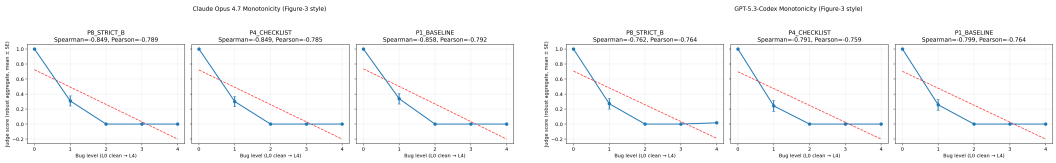
Judge–human agreement should be read against the reliability of the labels themselves. On the 72 analyzed pairs, inter-round agreement is moderate (Table 9), so validation-set judge–human Spearman values in the 0.48–0.56 range approach the available annotation ceiling on this panel rather than being "low absolute" values.

E.4 Cross-model validation status

The Claude Opus family run is a full-panel run of the same form as Codex and Qwen3-Coder: eight configurations (four Opus $\times$ prompts, cross-family `gpt-4o-mini` $\times$ two prompts, edit/random controls), repeated-judge aggregation, identical label views. Early Claude 3.x Sonnet scatterplots (Appendix H) were small-file protocol iterations used to stabilize prompts and ladders; they are not behind any headline statistic.

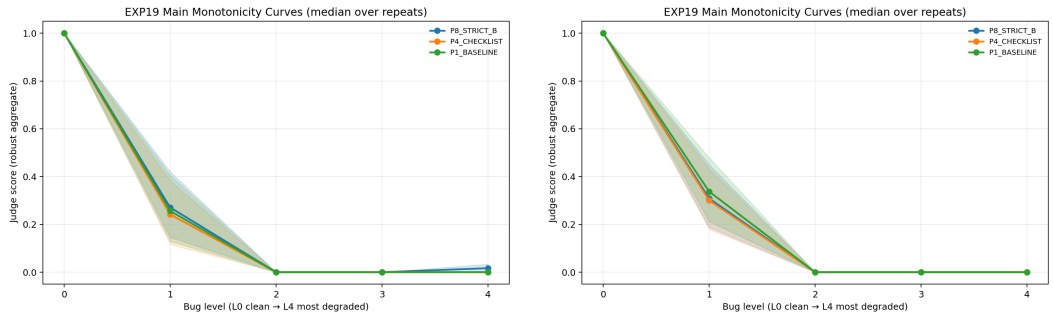
F Threshold Calibration

A threshold t converts a continuous judge score into a yes/no decision; flip rate measures how often repeated judge calls on the same item disagree after thresholding. Across $t \in$



(a) Claude Opus 4.7 monotonicity (mean ± SE). (b) GPT-5.3-Codex monotonicity (mean ± SE).

Figure 6: **Model-matched monotonicity diagnostics.** Both judge families show strong monotone score degradation as bug intensity increases under the same plotting protocol.



(a) Codex low-variance monotonicity curves. (b) Opus 4.7 low-variance monotonicity curves.

Figure 7: **Low-variance monotonicity trends across model families.**

$\{0.60, 0.65, 0.70, 0.75, 0.80, 0.85\}$, ranking behavior is unchanged ($\rho_{\text{bug}} = -0.790$ throughout), while stricter thresholds modestly reduce flip rate at some coverage cost; $t = 0.85$ gives the lowest observed flip rate with pass rate near 0.5.

G Downstream VeriBench Agent Ranking

As a portability sanity check outside the bridge panel, a property-checked judge (Claude 3.7 as both agent model and judge; 5 scoring calls per pair; integer rubric $\{0, \ldots, 10\}$ normalized by 10) ranks agentic theorem-generation methods in the expected direction:

H Judge Behavioral Diagnostics: Additional Plots

Early protocol-iteration scatterplots for the probe suite: (P1) identity, (P2) cumulative bug injection, (P3) removal of specification items. The bug plot under an early theorem-equivalence prompt was noisier, motivating the simplified rubric used in all main-paper results; these plots illustrate qualitative monotone behavior and are not behind headline statistics.

I Judge Prompt Templates

The primary Codex / Opus / Qwen comparisons use the prompt variants `P1_BASELINE`, `P4_CHECKLIST`, and `P8_STRICT_B`; the `BAD_STRINGY` lexical baseline is a negative control.

`P1_BASELINE`

```
You are an expert in Lean 4 formal verification.
Score how well the GENERATED THEOREM captures the correctness properties
of the REFERENCE IMPLEMENTATION. Score from 0.0 to 1.0.

- 1.0: Semantically equivalent to what the reference guarantees
- 0.7-0.9: Captures most properties, misses minor ones
- 0.4-0.6: Captures some properties, has meaningful gaps
- 0.1-0.3: Technically valid Lean but misses core properties
- 0.0: Trivially true (e.g., `True`) or incorrect
```

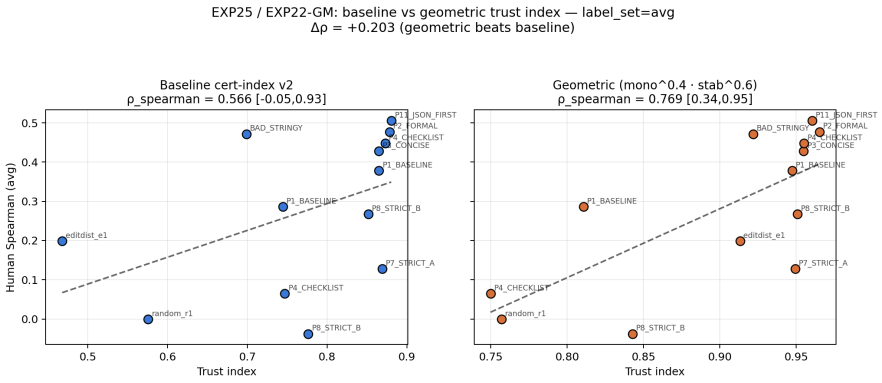
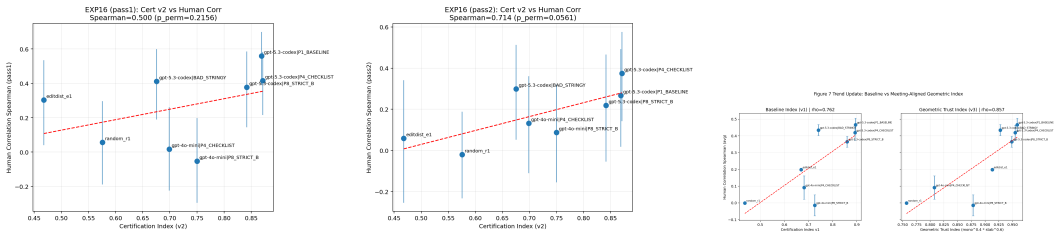


Figure 8: **Expanded 13-judge panel trend (avg validation labels).** The geometric aggregate retains strong alignment with TI_{m_1} at larger judge-panel size ($\rho = 0.769$); view-specific values are strict $\rho = 0.764$, lenient $\rho = 0.753$.



(a) Judge-level trend (strict) (b) Judge-level trend (lenient) (c) Judge-level trend (avg)

Figure 9: **Judge-level aggregate-to-criterion trend across label views.** Higher aggregate property satisfaction is associated with stronger human agreement in every view.

```
REFERENCE IMPLEMENTATION:
{reference_impl}

GENERATED THEOREM:
{generated_theorem}

Respond ONLY with valid JSON, no markdown:
{"score": <float 0.0-1.0>,
 "reasoning": "<one sentence max>"}
```

P4_CHECKLIST

```
Evaluate this in order:
1) What properties does REFERENCE IMPLEMENTATION guarantee?
2) Which are preserved by GENERATED THEOREM?
3) Are there missing core guarantees or trivialization?
4) Map to score in [0.0, 1.0]:
   1.0 equivalent; 0.7-0.9 minor gaps;
   0.4-0.6 meaningful gaps; 0.1-0.3 weak;
   0.0 trivial/incorrect.

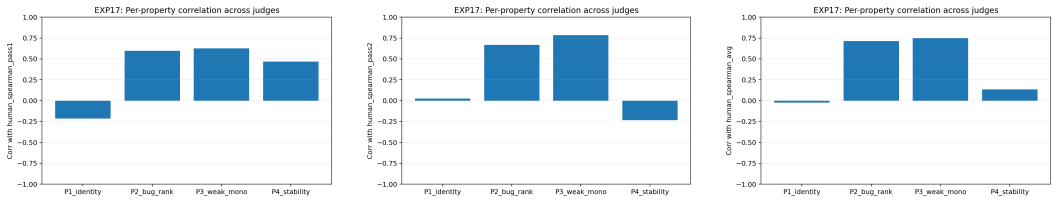
REFERENCE IMPLEMENTATION:
{reference_impl}

GENERATED THEOREM:
{generated_theorem}

Return JSON only:
{"score": <float 0.0-1.0>,
 "reasoning": "<one sentence>"}
```

P8_STRICT_B

```
Evaluate conservatively. If uncertain, choose the lower score.
```



(a) Per-property trend (strict) (b) Per-property trend (lenient) (c) Per-property trend (avg)

Figure 10: **Property-wise diagnostics.** P3 monotonicity is consistently the strongest predictor across label views; P2 second; P1/P4 weaker standalone at this scale.

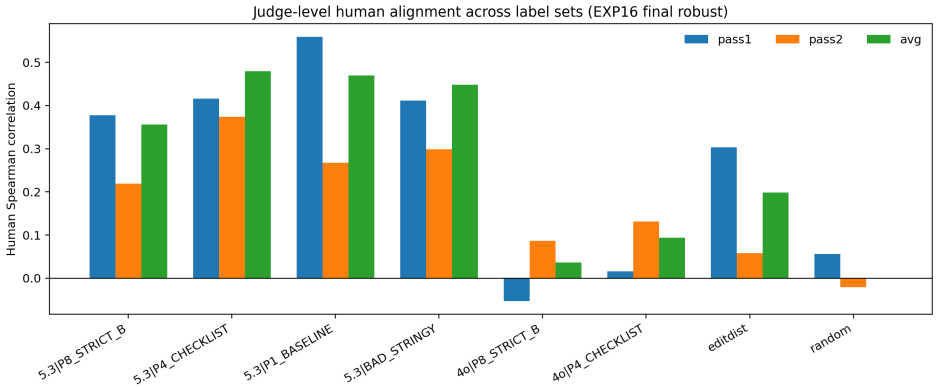


Figure 11: **Judge-wise human agreement across label views.** GPT-5.3-Codex variants remain above weaker baselines on most views; view-to-view variation motivates reporting all three.

```
Use [0.0,1.0]:
1.0 equivalent semantics
0.7-0.9 mostly right, minor misses
0.4-0.6 substantial omissions
0.1-0.3 weak relation
0.0 trivial/incorrect

REFERENCE IMPLEMENTATION:
{reference_impl}

GENERATED THEOREM:
{generated_theorem}

Return valid JSON only:
{"score": <float>,
 "reasoning": "<one sentence>"}
```

`BAD_STRINGY` negative control

```
You are a text similarity grader.
Compare generated theorem text to the reference implementation text.
Do NOT evaluate semantic correctness or theorem meaning.
Score ONLY based on superficial lexical/format overlap:
- high score if tokens/phrases look similar
- low score if wording/structure differs
Reference implementation:
{reference_impl}

Generated theorem:
{generated_theorem}

Return ONLY a JSON object:
{"score": <float 0.0 to 1.0>,
 "reasoning": "<short reason>"}
```

Label view	v1 Spearman	v2 Spearman
strict	0.762 ($p = 0.0372$, CI $[-0.139, 1.000]$)	0.524 ($p = 0.1959$, CI $[-0.400, 0.975]$)
lenient	0.810 ($p = 0.0222$, CI $[0.291, 1.000]$)	0.571 ($p = 0.1477$, CI $[-0.231, 0.975]$)
avg	0.762 ($p = 0.0402$, CI $[0.103, 1.000]$)	0.476 ($p = 0.2426$, CI $[-0.425, 1.000]$)

Table 7: **Higher-repeat rerun (9 repeats, median aggregation).** The $v1$ trend is preserved; $v2$ remains more sensitive at $n_{\text{judges}} = 8$.

Label view	ρ_s	r	τ	MAE
strict	0.171 ($p = 0.151$)	0.218 ($p = 0.0656$)	0.155 ($p = 0.144$)	1.049
lenient	0.388 ($p = 7.58 \times 10^{-4}$)	0.427 ($p = 1.87 \times 10^{-4}$)	0.343 ($p = 9.40 \times 10^{-4}$)	1.333

Table 8: **Naive blind baseline vs. validation labels (0–5 scale).** Included as a negative control; not used for ranking claims.

J Why Judge Scores Are Specification-Relative

A perfect judge score cannot guarantee absolute program correctness; the obstacle is computability, not benchmark design. Rice's theorem states that any non-trivial semantic property of programs in a Turing-complete language is undecidable (Rice, 1953): if a recursively enumerable gold standard captured every true semantic property of a program, it would yield a decision procedure contradicting the theorem. Gold references are therefore necessarily a curated, finite set of important specifications, and evaluations are *specification-relative*: a maximal score denotes close agreement with the audited gold formalization and rubric-conditioned human ratings, not a universal guarantee beyond that reference.

K Related Work (Continued)

Autoformalization. Wang et al. (2024a) present TheoremLlama, enhancing LLM translation into Lean 4 via NL–FL bootstrapping and iterative proof synthesis over 100K Mathlib4-derived examples, surpassing GPT-4 by over ten points on MiniF2F-Valid/Test.

Techniques for code verification. ImProver (Ahuja et al., 2024) introduces a Lean-aware Chain-of-States loop with retrieval and best-of-n sampling to rewrite formal proofs with improved properties. Clover (Sun et al., 2024) closes the loop between code, docstrings, and formal annotations with reconstruction-based consistency checks plus SMT verification, accepting 87% of correct solutions and rejecting 100% of flawed ones on CloverBench. Zhou et al. (2025) improve general-purpose graders with privileged information (gold solutions, rubrics, annotations), reaching expert-level grading on difficult benchmarks.

Agentic frameworks and tooling. Trace (Cheng et al., 2024) treats execution traces as gradients for generative optimization of full workflows; DSPy (Khattab et al., 2023) compiles declarative LLM pipelines with automatic prompt bootstrapping; Pantograph (Aniva et al., 2024) exposes Lean 4 proof states and tactics programmatically for learning agents.

Scalable oversight with LLM judges. Multi-agent oversight frameworks such as debate and consultancy use weaker judges to supervise stronger agents (Kenton et al., 2024; Chan et al., 2024); debate improves truth-tracking in math and code tasks. Self-critique frameworks such as SCRIT (Tang et al., 2025) bootstrap judgment skill without stronger supervisors. Validity-centered evaluation (Salaudeen et al., 2025) distinguishes measurement, claims, and evidence — construct, content, and criterion validity — and provides the reporting blueprint our construct/criterion/instrument split follows.

Robust evaluation without ground truth. Robertson & Koyejo (2025) model the overseer as an agent and propose information-theoretic, incentive-compatible queries (bounded f-divergences such as TVD-MI) with polynomial robustness to adversarial strategies. Peer-prediction consensus

Quantity	Value	Notes
Inter-round Spearman (strict vs lenient)	0.249	95% CI = $[-0.015, 0.489]$
Inter-round Pearson (strict vs lenient)	0.375	95% CI = $[0.076, 0.604]$
Inter-round Kendall τ	0.221	analyzed $n = 72$
Inter-round QWK	0.372	95% CI = $[0.073, 0.581]$
Inter-round MAE	0.771	analyzed $n = 72$, 0–5 rubric, mean-of-duplicates
Inter-round MSE	1.122	analyzed $n = 72$, 0–5 rubric, mean-of-duplicates
Best validation-set Spearman (avg labels)	0.480	best single judge, avg view
Best validation-set Spearman (strict labels)	0.559	best single judge, strict view

Table 9: **Single-rater rubric-stability context (analyzed $n = 72$).** All inter-round statistics use one duplicate-handling convention (mean-of-duplicates); an earlier version mixed conventions within this table (MAE 0.778, MSE 1.139), corrected here. With one annotator, these are rubric-stability figures, not inter-rater agreement.

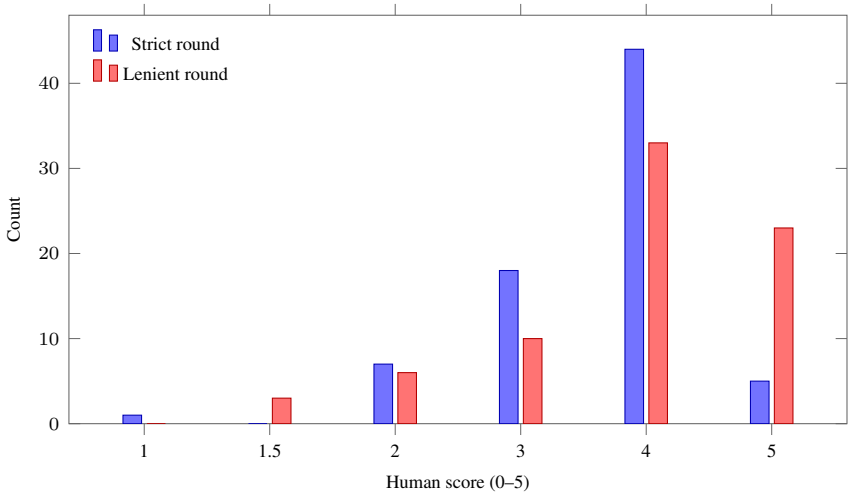


Figure 12: **Human-score dispersion across rubric rounds (75 rated pairs, 0–5 scale).** Counts are over the full 75 rated pairs (72 enter the analyses). The three 1.5 scores in the lenient round arise from duplicated legacy rows for one problem (scored 1 and 2, averaged before plotting); these duplicates are unrelated to the 3 pairs dropped from analysis, which fall out on a stale filename (Appendix A). The visible shift is consistent with the MAE/MSE statistics in Table 9.

certifies trust by asking whether multiple judges' outputs share information about a source — a population-level protocol. Behavioral-property screening is complementary: it evaluates a single judge in isolation, deployable when consensus is unavailable (single-judge inference, novel domains, sparse panels), and our validation protocol amortizes the human-annotation cost that consensus mechanisms also incur.

Consensus aggregation vs. property screening. Denisov-Blanch et al. (2026) stress-test five inference-time aggregation strategies across benchmarks and model scales; even at 25× single-sample cost, no aggregator consistently beats a single sample in verifier-absent domains, and several amplify shared misconceptions rooted in structurally correlated LLM errors (in MATH plurality-wrong cases the most common incorrect answer accounts for 65–87% of wrong responses; models agree at $\kappa \approx 0.35$ even on random-ASCII controls). Their diagnosis — consensus tracks what the crowd will say, not what is true — sharpens the case for property-based, single-judge screening whenever a verifier or calibrated reference exists; our work sits on the verifier-present side of that boundary. For oracle-free extensions, TVD-MI-style cross-judge information is a robustness diagnostic, not a truth proxy.

Specializing to a test set vs. screening across configurations. The *Specialist* method (Finkelstein et al., 2025) deliberately overfits a prompted autorater to a single benchmark using same-source historical ratings as demonstrations, with large gains on WMT MQM tasks. Both lines amortize the

Judge setup	Scale	Main	Role
GPT-5.3-Codex prompts (P8/P4/P1)	$n{=}8$	Yes	Primary judges
GPT-5.3-Codex BAD_STRINGY	$n{=}8$	Yes	Same-model stress
Qwen3-Coder prompts	$n{=}8$	Yes	Open-weights
Qwen3-Coder BAD_STRINGY	$n{=}8$	Yes	Open-weights stress
Claude Opus 4.x + controls	$n{=}8$	Yes	Third cross-model family
GPT-4o-mini prompts	$n{=}8$	Yes	Lightweight baseline
Edit-distance + random	$n{=}8$	Yes	Non-LLM controls
Claude 3 Sonnet (narrow slice)	Small	No	Prompt diagnostics

Table 10: **Cross-model validation status.** Headline bridge claims use full eight-configuration panels on the analyzed rows; exploratory Sonnet figures are withheld from headline statistics.

Threshold	Flip rate (95% CI)	Pass rate (95% CI)
0.60	0.058 [0.048, 0.069]	0.583 [0.559, 0.608]
0.70	0.049 [0.040, 0.059]	0.559 [0.537, 0.581]
0.80	0.051 [0.042, 0.061]	0.544 [0.523, 0.566]
0.85	**0.046** [**0.037**, **0.057**]	0.536 [0.515, 0.557]

Table 11: **Threshold calibration summary.** Modest, consistent reduction in decision flips at stricter thresholds, with moderate coverage loss; ranking unchanged across thresholds.

expensive part of human evaluation but along orthogonal axes: Specialist amortizes *within* a test set (and, as its authors show, specializes to the individual rater), while property screening amortizes *across judge configurations* with no human labels on perturbed variants. Their single-rater specialization finding echoes our single-annotator ceiling; both motivate multi-rater, protocol-consistent held-out designs.

Formal methods as oversight. Zhou et al. (2024) filter autoformalized solutions via proof checking, outperforming majority voting on GSM8K; SAFE (Liu et al., 2025) verifies intermediate steps in Lean 4 and introduces FormalStep. Ospanov et al. (2023) integrate Lean feedback into repair cycles, boosting proof success on miniF2F; Miranda et al. (2026) extend the paradigm to program verification, where self-debugging agents quadruple success over one-shot prompting.

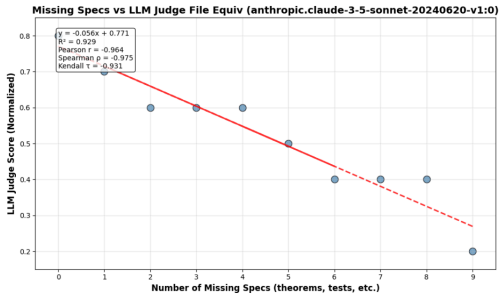
Reference-free judging of autoformalization. Zhang et al. (2025) score formalizations without a gold reference via an ensemble with a fine-grained taxonomy — scalable when golds are unavailable, but validated against subjective human preference rather than machine-checkable equivalence, and motivated by a small non-random sample of "potentially incorrect" formalizations pre-selected by an uncertified LLM. Benchmark noise rarely alters relative model rankings because errors correlate across systems (Salaudeen & Hardt, 2024), so a small number of flawed gold examples does not invalidate comparative evaluation; we proceed from kernel-checked gold references and emphasize calibration evidence for pairwise scores.

Verified-coding benchmarks and specification generation. Bursuc et al. (2025) presume well-formed specifications and evaluate end-to-end verification under them (Dafny, Verus/Rust, Lean). Impactful problems often lack both complete specifications and ground-truth answers; VERIBENCH targets the specification-*generation* gap, and the Unsolved Questions paradigm (Nie et al., 2025) extends evaluation to specless, answerless settings via validator pipelines. Together these shift evaluation from "solve what is specified" to "specify, calibrate, then solve"; our contribution is the validation layer for the semantic judges that grading generated specifications requires.

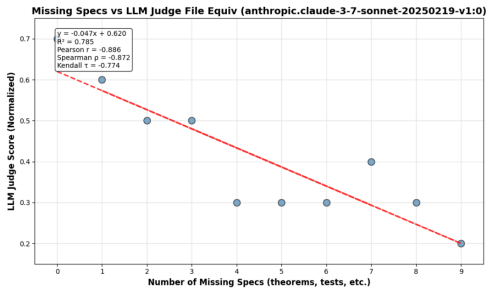
HCI for oversight. HCI reframes oversight as calibrated trust: interfaces that expose process and embed lightweight verification rather than one-shot answers (Bach et al., 2022; Ding et al., 2025; Wang et al., 2024b). Interaction loops that make correctness checkable during generation — test-elicitation (Naik et al., 2024), explain-then-correct (Yan et al., 2024), pair debugging (Ma et al., 2024), exposed multi-step pipelines (Wu et al., 2022), and source-grounded verification (Hennigen et al., 2024) — shift user trust from fluency to verifiable progress. Two patterns port directly to formal specs: counterfactual spec probes (remove a postcondition or inject a bug, display the

Agent	Easy	CS	Real	HE	Sec	Avg
Baseline	0.580	0.690	0.472	0.410	0.347	**0.470**
DSPy ReAct	0.659	0.754	0.580	0.650	0.454	**0.615**
TRACE+	0.342	0.680	0.592	0.573	0.411	**0.477**
TRACE++	0.620	0.756	0.504	0.645	0.432	**0.588**

Table 12: **Theorem quality scores on VERIBENCH** (average normalized scores by subset). Feedback-driven methods outscore simple prompting, led by DSPy ReAct (Khattab et al., 2023; Cheng et al., 2024; Chen et al., 2021). This configuration is outside the eight-judge bridge panel; the result is a portability sanity check, not independent proof of judge correctness.



(a) Claude 3.5 Sonnet v1: strong monotonic decrease ($y = -0.056x + 0.771$, $R^2 = 0.929$, Pearson $r = -0.964$, Spearman $\rho = -0.975$).



(b) Claude 3.7 Sonnet v1: clear negative trend with minor non-monotonicity ($y = -0.047x + 0.620$, $R^2 = 0.785$, Pearson $r = -0.886$, Spearman $\rho = -0.872$).

Figure 13: **(P3) Monotonicity w.r.t. missing specifications.** Scores decrease as more theorems/tests are withheld; 3.7's small non-monotonicity motivated the simplified rubric used in the main results.

score impact) and stepwise proof oversight (surface intermediate goals, mark unverified suggestions); both interface naturally with a graded, semantics-sensitive evaluator and a judge-card report in the validity-centered sense of Salaudeen et al. (2025).

Formal-verification context. With a proof assistant, oversight decomposes: the kernel deterministically checks every proof step, so the human verifies a much smaller artifact — the statement, definitions, and assumptions. Lean 4 lets runnable code, specifications, and machine-checked proofs live in one dependently-typed language with mature tooling (`mathlib4`, metaprogramming, Pantograph, Aesop), which is why VERIBENCH adopts it; its guarantees remain static and specification-relative, and theoretical limits (Rice, halting, incompleteness) bound what the verifier can settle — not the oversight claim that humans are best deployed on asking the right question. Alternatives trade differently: Dafny's SMT-centered workflow offers push-button automation but not Lean's value-indexed, higher-order specifications; TLA+ excels at protocol specification but is not executable; Verus verifies Rust with a larger trusted surface than Lean's small kernel.

Hybrid TVD-MI protocols. Where gold specifications exist, calibrating TVD-MI-style information scores against known gold outcomes (compile success, unit tests, post-condition proofs) could yield composite metrics that preserve gaming-resistance while adding calibrated correctness probabilities — a practical bridge between unsupervised robustness and supervised precision (Robertson & Koyejo, 2025).

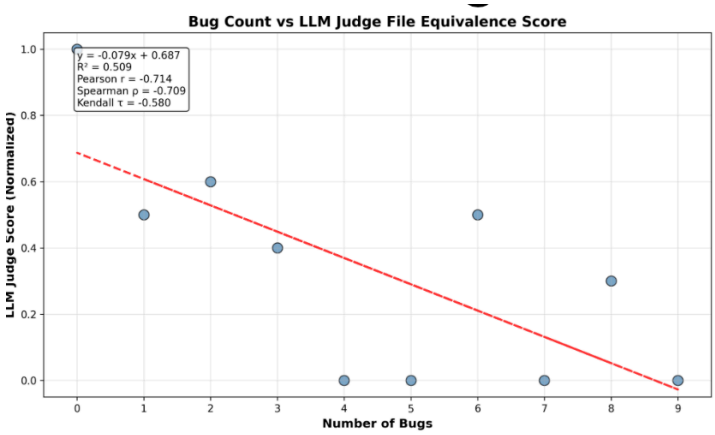


Figure 14: **(P2) Monotonicity w.r.t. cumulative bugs (Claude 3.5 Sonnet).** Scores drop as more semantic bugs are injected (Pearson $r = -0.714$, Spearman $\rho = -0.709$, $R^2 = 0.509$); the noise under the initial prompt led to the simplified final rubric.

L Pre-Registration Summaries

Two held-out screens were pre-registered against the revised-protocol criterion (Section 7); their documents (candidates, decision thresholds, analysis seeds, success bars, committed before new judge scores existed) ship in the supplementary archive. **Screen 1** (14 configurations): registered outcome *underpowered*. **Screen 2** (32 configurations): registered outcome *negative* — no candidate met the success bar (strongest $|\rho| = 0.39$, Holm-corrected $p = 0.25$; leave-one-out selection transfer $\rho = 0.24$, permutation $p = 0.49$). The registered next analyses are: re-anchored calibration under one rubric end-to-end; expansion toward $\approx$40 judge configurations; the δ-measurement collection for Route B (Appendix B); and decision-utility measurement at a fixed false-pass tolerance (Table 2).